Run 1 vs Run 2: Comparison & Final Capstone Report

AI-Powered Engineering Document Quality Assurance Agent | Fictional training example

1. Project objective

Evaluate a six-step fictional centrifugal pump maintenance procedure for document-quality gaps, unclear instructions, missing acceptance criteria, and required verification sources. The agent is expected to provide evidence-based findings and recommendations without inventing equipment-specific requirements.

2. Run comparison

Review dimension	Run 1	Run 2	Assessment
Number of findings	5 findings	7 findings	Run 2 covers more distinct omissions.
Evidence	Step-specific evidence for five findings.	Step references and whole-procedure finding.	Run 2 has broader coverage.
Severity	High / Medium	High / Medium / Low	Both label severity as preliminary; Run 2 has a wider range.
Corrective actions	General recommendations.	More specific actions linked to verification sources.	Run 2 is more actionable.
Verification sources	Mentioned in recommendations.	Dedicated source column for each finding.	Run 2 improves auditability.
Checklist / traceability	Checklist summary and brief work log.	Expanded checklist, assumptions, unanswered questions, and work log.	Run 2 documents review method and limitations more clearly.

3. Specific improvements in Run 2

F-02 — Safe access: The report explicitly identifies that the procedure says to stop the pump and open it without defining safe-to-open conditions.

F-06 — Maintenance records: The report specifies example traceability fields such as equipment identification, date, work performed, parts used, findings, corrective actions, responsible personnel, and applicable test results.

F-07 — PPE and risk assessment: The report adds a whole-procedure finding for missing PPE and risk-assessment references.

Better verification planning: Run 2 provides a verification-source column and a list of unanswered questions that require approved documents or qualified personnel.

4. What remains unresolved

Neither run verifies the procedure against an OEM manual, approved site isolation procedure, site risk assessment, equipment configuration, or formal risk matrix. Severity ratings are preliminary. The reports identify documentation gaps; they do not prove that an actual pump is unsafe or that a specific technical requirement has been violated. Do not use this fictional report as an operational maintenance instruction.

5. Feedback-loop diagnosis

Lever	Run 1 limitation	Change reflected in Run 2
Context	Limited checklist-level traceability and fewer explicitly separated gaps.	More complete review areas and explicit unanswered questions; however, actual OEM/site sources remain unavailable.
Instructions	Output structure was less detailed for evidence, impacts, actions, and verification.	A more structured findings table with severity, impact, corrective action, and verification source.
Workspace	No external/OEM documents were available.	This remains unresolved; Run 2 states external/OEM verification was not performed.

6. Final conclusion

Run 2 is the stronger review artifact because it identifies seven findings, separates potential impact from corrective action, names verification sources, and documents assumptions and unresolved questions. The iteration improved the report's completeness and auditability. The key remaining limitation is the absence of approved technical source documents; the agent appropriately leaves site-specific questions unresolved rather than inventing values or requirements.

7. Suggested GitHub repository structure

```
engineering-document-qa-agent/
■■■ README.md
■■■ reports/
■ ■■■ Run_1_Engineering_Document_QA_Report.pdf
■ ■■■ Run_2_Engineering_Maintenance_Procedure_Quality_Review.pdf
■ ■■■ Run_1_vs_Run_2_Comparison_and_Final_Capstone_Report.pdf
■■■ docs/
■■■ review-checklist.md
```

8. GitHub README draft

Project overview
This project demonstrates a human-supervised AI agent workflow for reviewing engineering maintenance documentation. A fictional six-step centrifugal pump procedure was reviewed in two iterations. Run 1 established a baseline; Run 2 expanded finding coverage, improved evidence traceability, added verification sources, and documented assumptions and unresolved questions.

Workflow
1. Provide a fictional procedure and quality checklist.
2. Ask the agent to identify evidence-based documentation gaps.
3. Review Run 1 and diagnose weaknesses.
4. Improve the instruction structure and review criteria.
5. Run again and compare findings, traceability, and limitations.
6. Keep a human reviewer responsible for final acceptance.

Key learning
Better agent results require more than rerunning the same prompt. The reviewer should diagnose whether the issue is context, workspace/tool access, or instructions, then make deliberate changes and compare the new trajectory and output.

Safety and limitations
The example is fictional and for training only. No OEM documentation, site-approved procedures, or formal risk assessment was verified. It is not approved for real maintenance use; all operational requirements must be confirmed by qualified personnel against approved sources.

9. Final submission checklist

■ Save both run reports and this comparison PDF in the repository.
■ Add README.md with the project purpose, workflow, comparison, and limitations.
■ If permitted, include the anonymized/fictitious source procedure and checklist used.
■ Record the actual prompt changes between Run 1 and Run 2; do not claim changes you did not make.
■ Confirm at least two full runs were completed.
■ Before the defense interview, be ready to explain what improved, what remained unresolved, and why.

Prepared as a portfolio and certification-support document from the two supplied reports. Confirm that this accurately reflects the work you personally completed before submitting it.